

Microbiology Labs





About the Course

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Microbiology



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
12	Microbiology Labs 1 time/week 60 min	Illustrated Guide to Home Biology Experiments

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Microbiology				
Microbiology Lessons Nature Walks & Scouting: Grades 9-12	Microbiology Lessons	Microbiology Lessons	Microbiology Lessons Nature Notebook: Grades 9-12	Microbiology Lessons Microbiology Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Microbiology Labs



Illustrated Guide to Home Biology Experiments



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

(No Quick Links Assigned)

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or scan the QR
code for links.](#)



Microbiology Labs

How To Approach Labs



Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

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Term 1

WEEK 1 60m Microbiology Labs - Lesson 1

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

Continue reading p.3-51. Then, skim through the rest of the book and plot out a rough plan of which experiments you'd like to explore. Order the needed supplies. Home Scientist has a kit that contains the basic supplies needed. You can add on bacterial cultures and prepared slides, if desired. If you order supplies elsewhere, you can still order the DVD with answers to the review questions in the book and high-resolution images as a separate item. As the introductory material in the book says, if you decide to do labs that require bacterial cultures or prepared slides of certain items that you cannot procure locally, you will need to order those items from Home Scientist ahead of time. They suggest that you could use the high-resolution images that come with their DVD kit instead of purchasing slides.

→ VIEW

∞ Link: Home Scientist

WEEK 2 60m Microbiology Labs - Lesson 2

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

Today, you will complete LAB I-1: Using a Microscope. Gather all of the needed materials. Prepare your Lab Notebook for the experiment. Read the lab carefully before beginning and adhere to all safety requirements. Record what you do and what you observe in your Lab Notebook. Then clean up and dispose of any materials as instructed.

Complete the lab write-up in your notebook, including the review questions. After you have finished your work, use the answers in the PDF that came with your supplies and see if there was anything significant that you did not understand. Use your Biology textbook from Grade 10 and the internet to learn more or aid in understanding terms and concepts.

You will complete your lab report following the book's guidelines. Choose 6 throughout the year to complete as full lab reports. Use what you have learned about writing lab reports. Use the guidelines for the rest of your reports.

WEEK 3 60m Microbiology Labs - Lesson 3

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 4 60m Microbiology Labs - Lesson 4

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 5 60m Microbiology Labs - Lesson 5

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

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Term 1		
WEEK 6	60m Microbiology Labs - Lesson 6 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 7	60m Microbiology Labs - Lesson 7 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 8	60m Microbiology Labs - Lesson 8 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 9	60m Microbiology Labs - Lesson 9 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 10	60m Microbiology Labs - Lesson 10 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 11	60m Microbiology Labs - Lesson 11 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 12	60m Microbiology Labs - Lesson 12 Exam Week → EXAMS Answer question(s) related to course.	

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Term 2

WEEK 1 60m Microbiology Labs - Lesson 13

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

Continue lab work, and write up your report after completing each lab.

WEEK 2 60m Microbiology Labs - Lesson 14

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 3 60m Microbiology Labs - Lesson 15

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 4 60m Microbiology Labs - Lesson 16

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 5 60m Microbiology Labs - Lesson 17

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 6 60m Microbiology Labs - Lesson 18

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 7 60m Microbiology Labs - Lesson 19

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 8 60m Microbiology Labs - Lesson 20

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 9 60m Microbiology Labs - Lesson 21

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 10 60m Microbiology Labs - Lesson 22

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

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Term 2

WEEK 11 60m Microbiology Labs - Lesson 23 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 12 60m Microbiology Labs - Lesson 24 Exam Week → EXAMS Answer question(s) related to course.	

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Term 3

WEEK 1 60m Microbiology Labs - Lesson 25

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

Continue lab work, writing up your report after completing each lab.

WEEK 2 60m Microbiology Labs - Lesson 26

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 3 60m Microbiology Labs - Lesson 27

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 4 60m Microbiology Labs - Lesson 28

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 5 60m Microbiology Labs - Lesson 29

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 6 60m Microbiology Labs - Lesson 30

Materials: Illustrated Guide to Home Biology Experiments

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WEEK 7 60m Microbiology Labs - Lesson 31

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 8 60m Microbiology Labs - Lesson 32

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 9 60m Microbiology Labs - Lesson 33

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

WEEK 10 60m Microbiology Labs - Lesson 34

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

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Term 3

WEEK 11 60m Microbiology Labs - Lesson 35 Materials: Illustrated Guide to Home Biology Experiments → LAB DAY	
WEEK 12 60m Microbiology Labs - Lesson 36 Exam Week → EXAMS Answer question(s) related to course.	